
Discrete Edit Adjoint Matching for Reward Fine-Tuning Variable-Length Discrete Generators

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Abstract

Reward fine-tuning of variable-length discrete generators requires controlling insertion, deletion, and substitution rates while preserving the underlying continuous-time dynamics. We introduce Discrete Edit Adjoint Matching (DEAM), which extends adjoint matching to edit CTMCs, chains on sequences of any length whose transitions are single edits, using the full path-space likelihood ratio and a learned scalar potential whose differences define log-rate corrections. We amortize it into a direct edit head, avoiding a potential evaluation at every one-edit successor during sampling. On an enumerable Edit Flow with exact values and adjoints, DEAM approaches the discretized exact-adjoint reference without oracle access and substantially outperforms non-oracle rate-matching and trajectory-credit baselines. The same controller transfers to small-molecule design: on a pretrained 96M-parameter any-length insertion diffusion over SAFE strings, DEAM matches A2D2’s reward at better synthetic accessibility, validity and diversity.

1 Introduction

Reward fine-tuning of a pretrained generative model can be posed as entropy-regularized stochastic control. Let $X = (X_t)_{t \in [0,1]}$ be a stochastic process on a discrete state space $\mathcal{X}$, let p^u denote its path law under model parameters u , and let p^{base} denote the pretrained path law. We solve:

$$\min_u \mathbb{E}_{X \sim p^u} [g(X_1)] + D_{\text{KL}}(p^u \| p^{\text{base}}), \quad (1)$$

with $g = -r/\alpha$ a terminal loss built from a reward $r : \mathcal{X} \rightarrow \mathbb{R}$ and temperature $\alpha > 0$. For a continuous-time Markov chain (CTMC) with off-diagonal base transition rates $u_t^{\text{base}}(y | x)$:

$$u_t^*(y | x) = u_t^{\text{base}}(y | x) e^{-V_t(y) + V_t(x)}, \quad V_t(x) := -\log \sum_z p_{1|t}^{\text{base}}(z | x) e^{-g(z)}, \quad (2)$$

where $p_{1|t}^{\text{base}}(z | x) := \Pr_{p^{\text{base}}}(X_1 = z | X_t = x)$ is the base completion probability [1, 5]. Every transition is its base rate scaled by an exponential value difference, so fine-tuning reduces to estimating V . We hold the source p_0 fixed rather than assuming a memoryless base schedule, so the optimum tilts the base law per initial state and p_1^* is the fixed-source tilt rather than the global terminal tilt of p_1^{base} (Appendix B). Discrete Adjoint Matching (DAM) makes this operational by constructing a sample-based estimator of the difference and matching the target.

We study this problem on edit CTMCs: chains on $\mathcal{X} := \bigcup_{n=0}^N \mathcal{T}^n$ for a finite vocabulary $\mathcal{T}$ and maximum length N , whose rate $u_t(y | x)$ is nonzero only when y is obtained from x by a single insertion, deletion, or substitution. The class spans architectures: Edit Flows [3] realize it with all three operations, and A2D2 [6] with any-length insertion diffusions. It is also the natural class for molecular design, where a molecule written as a SAFE or SMILES string has no fixed token count and improving drug-likeness or synthesizability means adding, removing and replacing fragments.

Reward fine-tuning over this class remains limited. A2D2 reweights insertion and unmasking objectives with a trajectory-level importance weight and LPDP [4] steers Edit Flows at inference time, but neither optimizes reward at training time over the full edit class. Existing training-time constructions do not transfer directly: DAM’s CTMC adjoint is implemented for masked diffusion, A2D2 requires an analytic insertion-unmasking interpolant, and the alignment paths used to train Edit Flows are not in general the endpoint-conditioned bridges of the learned CTMC.

We introduce Discrete Edit Adjoint Matching (DEAM), an adjoint method for edit CTMCs. (1) We estimate state values with the path-space likelihood ratio and induce edit-rate corrections through differences of a learned scalar potential, which we amortize into an edit head, reducing sampling to one network evaluation per step. (2) On an enumerable Edit Flow with exact values, DEAM reaches the reference without oracle access. (3) The same controller transfers to a pretrained any-length molecule generator, matching A2D2’s reward at better synthetic accessibility, validity and diversity.

2 Method

Path-space potential matching DEAM estimates V_t at sampled states from the exponential value difference in (2) and derives all transition tilts from a learned scalar potential. Although the marginal completion probability $p_{1|t}^u$ is intractable, the likelihood ratio of completion paths can be evaluated from rates along a trajectory. Let $X_{t:1} := (X_s)_{s \in [t,1]}$ have jumps at $t < \tau_1 < \dots < \tau_m \leq 1$, and define $\Lambda_t^v(x) := \sum_{y \neq x} v_t(y | x)$. The log Radon–Nikodym derivative is

$$W_{[t,1]}(X_{t:1}) := \log \frac{d p_{\cdot|X_t=x}^{\text{base}}(\cdot | X_t=x)}{d p^u(\cdot | X_t=x)}(X_{t:1}) = \sum_{j=1}^m \log \frac{u_{\tau_j}^{\text{base}}(X_{\tau_j} | X_{\tau_j^-})}{u_{\tau_j}(X_{\tau_j} | X_{\tau_j^-})} + \int_t^1 (\Lambda_\tau^u(X_\tau) - \Lambda_\tau^{\text{base}}(X_\tau)) d\tau. \quad (3)$$

Both terms are accumulated along the same model rollout, and since g depends only on the endpoint, $\mathbb{E}_{p^u}[e^{W_{[t,1]}(X_{t:1}) - g(X_1)} | X_t = x] = \mathbb{E}_{p^{\text{base}}}[e^{-g(X_1)} | X_t = x] = e^{-V_t(x)}$. In DAM’s masked-diffusion specialization, the model and base share the prescribed total jump rate, so the integral in (3) vanishes. Edit CTMCs learn unnormalized insertion, deletion and substitution rates, so it must be retained; dropping either term breaks the estimator by orders of magnitude (App.B.3).

Given K independent model completions with endpoints $Z^{(k)} := X_1^{(k)}$ and weights $W^{(k)} := W_{[t,1]}(X_{t:1}^{(k)})$, we estimate $\hat{V}_t(x)$ and learn $V_\theta : \mathcal{X} \times [0,1] \rightarrow \mathbb{R}$ by regressing onto it:

$$\hat{V}_t(x) = -\log \frac{1}{K} \sum_{k=1}^K e^{W^{(k)} - g(Z^{(k)})} \quad (4) \quad \mathcal{L}_V(\theta) = \mathbb{E}_{t, X_t \sim p_t^\theta} \left[(V_\theta(X_t, t) - \hat{V}_t(X_t))^2 \right] \quad (5)$$

where p_t^θ is the time- t marginal of the current CTMC, and induce the controlled rates through its finite differences,

$$u_t^\theta(y | x) = u_t^{\text{base}}(y | x) \exp(s_\theta(x, y, t)), \quad s_\theta(x, y, t) = V_\theta(x, t) - V_\theta(y, t), \quad (6)$$

The edit rates are therefore coupled through a common scalar potential rather than parametrized independently. Since the exact log-rate correction is a difference of state values, this enforces consistency across edits especially as corrections are antisymmetric on reversible edits. It also moves supervision from edges to states: one rollout target trains $V_t(x)$ and can be reused in every edit correction involving x .

Amortizing the potential Sampling from u^θ in (6) requires evaluating V_θ at the current state and every one-edit successor. We therefore learn a direct edit head s_ϕ that, from the current state x , emits corrections for all $y \in N(x)$ in one shared forward pass, with $N(x)$ the single-edit neighbours of x . This replaces one critic evaluation per successor by one network evaluation, while retaining an $O(|N(x)|)$ -dimensional output. The resulting rates are $u_t^\phi(y | x) = u_t^{\text{base}}(y | x) e^{s_\phi(x, y, t)}$. Its regression target is the potential difference of a delayed copy $\bar{\theta}$ of the critic, $\bar{s}(x, y, t) := V_{\bar{\theta}}(x, t) - V_{\bar{\theta}}(y, t)$ which is precisely the quantity s_θ induces in (6). Only M neighbours per visited state are supervised, drawn from a proposal $\rho(\cdot | x)$ on $N(x)$ and reweighted by $1/(M\rho(y | x))$, so that for any ρ the estimator is unbiased for the full generalized-KL sum over $N(x)$:

$$\mathcal{L}_s(\phi) = \mathbb{E} \left[\frac{1}{M} \sum_{y \sim \rho} \frac{u_t^{\text{base}}(y | x)}{\rho(y | x)} \left(e^{s_\phi(x, y, t)} - e^{\bar{s}(x, y, t)} s_\phi(x, y, t) \right) \right] \quad (7)$$

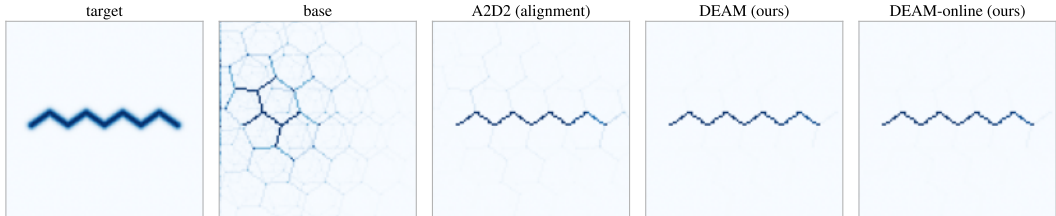


Figure 1: Seed-averaged terminal distributions at the matched 48-hour budget: each string’s curve drawn as ink weighted by its terminal probability. Steered arms are separated by sequence length (Appendix Figure 4) and path KL (Table 1).

We take ρ to be an equal mixture of the normalized base and actor jump distributions. An on-policy proposal concentrates on edits that already carry high rate and under-samples those the controller must suppress, leaving their rates near the base value. Because the actor is the behaviour policy, actor and critic co-adapt. Algorithm 1 summarizes the resulting loop.

3 Experiments

Setup. We first study an enumerable Edit Flow over binary strings, $\mathcal{X} = \bigcup_{n=0}^{12} \{0, 1\}^n$, where each token is a deterministic turtle turn, so a length- n string renders as an n -segment planar curve. The target is the curve of the alternating length-eight string. The reward $r(x) \in [0, 1]$ multiplies two Gaussian-kernel coverage scores penalizing missing and extraneous segments, so it depends on token content and sequence length alike. At $t = 0$ the length is uniform on $\{0, \dots, 12\}$ and the string uniform given its length; base insertion, deletion and substitution rates are 1.0, 0.4, 0.8, and $g = -r/0.025$ (Appendix B).

Implementation and comparisons. The state space contains 8191 strings with maximum edit degree 46, so base values and the optimal rate are exact by sparse matrix-exponential products. All learned rates are piecewise constant over 20 time bins, so the exact adjoint at the bin midpoints defines a discretized reference at terminal KL 0.155. All neural arms share state features and MLP trunk. DAM uses the same rollout adjoint targets but predicts independent per-edge corrections, its generalized-KL loss summed over all admissible edits to remove edge-sampling variance. A2D2-Align applies A2D2’s trajectory weighting to the alignment interpolant used in Edit Flow training, which is not in general the bridge of our base CTMC; A2D2-Bridge supplies that bridge exactly (Appendix B.2). A2D2-Bridge and exact-adjoint DAM are oracle diagnostics, and all arms are matched to a 48-hour wall-clock budget per seed.

Results. At matched budget DEAM reaches terminal KL 0.114 ± 0.009 , below the discretized exact-adjoint reference and both non-oracle baselines, at path KL 1.04 against 17.8 and 38.5 (Table 1); steering changes sequence length as well as token content (Appendix Figure 4). A2D2-Align improves the terminal marginal while raising path KL above the base model (17.8 vs. 11.3), whereas the same credit under the exact CTMC bridge achieves 0.195 ± 0.009 at path KL 0.34: endpoint gains can mask dynamical disagreement arising from the conditional path rather than from trajectory-level weighting. DAM’s path KL stays high because its free per-edge head reaches the endpoint by running $3\times$ the optimum’s edit rate (Appendix C.2). The amortized actor matches the critic at lower path KL (0.54 vs. 1.04) while replacing successor-wise critic evaluation by one shared actor pass, and is insensitive to $M \in \{1, 2, 4, 8, 16\}$ (Appendix C.3). The ablation in Appendix C.4 varies the parametrization and the loss independently: the free per-edge head reaches 0.287 under rate-space gKL and 1.037 under log-space regression, and the potential diverges under gKL, while their combination reaches 0.114.

Scaling to a pretrained generator We then fine-tune A2D2’s pretrained any-length insertion MDM over SAFE molecules [6] (96M parameters, vocabulary 1882, length to 256): an edit CTMC supported on insertions, at edit degree up to $\approx 5 \times 10^5$, which tests whether a scalar potential still resolves per-edit differences at that degree. The setting favours A2D2, whose training interpolant is the endpoint-conditioned bridge. We match its objective $r = \text{QED} + 1/(1 + \text{SA})$ and temperature.

Table 1: Terminal and path KL at matched 48-hour training budget per seed (5 seeds for the non-oracle arms, 3 for the oracle rows; median over the final five evaluations per seed, then mean $\pm$ sd).

Method	Adjoint / credit source	$D_{\text{KL}}(p_1^* \ p_1^u)$	path KL
Base model	—	8.413	11.3
DAM, all edges	rollout adjoint	0.287 ± 0.029	38.5
A2D2	alignment coupling	0.939 ± 0.012	17.8
DEAM (ours)	rollout adjoint, potential	0.114 ± 0.009	1.04
DEAM-online (ours)	+ amortized actor	0.133 ± 0.006	0.54
A2D2	<i>oracle CTMC bridge</i>	0.195 ± 0.009	0.34
DAM, all edges	<i>exact adjoint</i>	0.160 ± 0.000	0.05
Reference (u^*)	exact	0.155	—

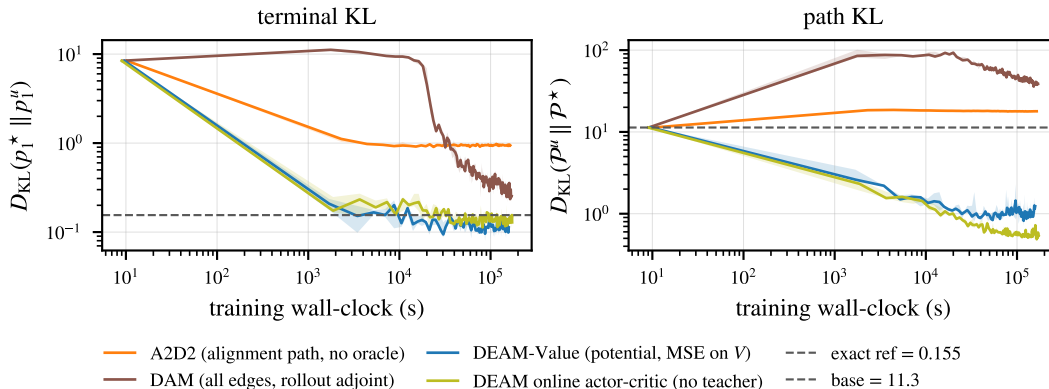


Figure 2: Terminal KL (left) and path KL (right) versus training wall-clock, median over five seeds. Both DEAM variants fall below the discretized exact-adjoint reference on terminal KL. The rollout axis is given in Appendix C Figure 3.

The backbone stays frozen; DEAM trains a summed per-position potential and the amortized edit head ($\approx 1\text{M}$ parameters) in a single loop from random initialization. Value targets come from base-model completions, keeping the path-ratio correction identically zero; DAM’s importance-weighted estimator degenerates at this temperature (Appendix D).

Table 2: Drug-like small molecule design, 1000 samples $\times$ 3 seeds. A2D2 numbers from [6].

Method	QED ($\uparrow$)	SA ($\downarrow$)
Pre-trained (any-length)	0.653 ± 0.003	3.472 ± 0.040
A2D2	0.762 ± 0.004	2.870 ± 0.018
DEAM (ours)	0.717 ± 0.005	2.807 ± 0.015

DEAM reaches comparable reward on A2D2’s own benchmark while improving synthetic accessibility (Table 2), and does so at higher validity, uniqueness and diversity. QED-only is ablated, it recovers A2D2’s QED but degrades SA (Appendix D).

Conclusion We introduced DEAM, adjoint matching for edit CTMCs through a scalar potential whose state differences set the log-rate corrections, amortized into an edit head costing one network pass per sampling step. It enforces consistency across edits and attains the best non-oracle terminal and path KL on an enumerable Edit Flow, and transfers to a pretrained molecule generator at edit degree up to $\approx 5 \times 10^5$ where DEAM matches A2D2’s reward at better synthetic accessibility and diversity.

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A Training loop

Algorithm 1 DEAM training

Require: base rates u^{base} , terminal loss g , critic V_θ , actor s_ϕ , completions K , supervised edits M

- 1: **for** each iteration **do**
- 2: $\bar{u} \leftarrow \text{stopgrad}(u^\phi)$ ▷ the actor is the behaviour policy
- 3: **for** each batch element **do**
- 4: sample $t \sim \mathcal{U}[0, 1]$ and simulate $\bar{u}$ to obtain X_t
- 5: sample K completions from (X_t, t) under $\bar{u}$, accumulating weights using (3)
- 6: compute $\hat{V}_t(X_t)$ using (4)
- 7: draw M neighbours $y \sim \rho(\cdot | X_t)$ with targets $\bar{s}(X_t, y, t) = V_\theta(X_t, t) - V_{\bar{\theta}}(y, t)$
- 8: **end for**
- 9: update θ, ϕ on the empirical versions of (5) and (7)
- 10: periodically $\bar{\theta} \leftarrow \theta$
- 11: **end for**

B Toy experiment details

B.1 Experimental Setup

Reward. Each binary string is rendered by a unit-step turtle with initial heading 35° ; tokens 0 and 1 turn the heading by -70° and $+70^\circ$, respectively. The target is the curve traced by the alternating length-eight string. Let $\mathcal{P}_T$ and $\mathcal{P}_x$ denote points sampled along the target and rendered curves, and let $d(p, \mathcal{S})$ be the Euclidean distance from a point p to the closest segment in segment set $\mathcal{S}$. We use the symmetric coverage reward

$$r(x) = \left[\frac{1}{|\mathcal{P}_T|} \sum_{p \in \mathcal{P}_T} \exp\left(-\frac{d(p, \mathcal{S}_x)^2}{2\sigma^2}\right) \right] \left[\frac{1}{|\mathcal{P}_x|} \sum_{p \in \mathcal{P}_x} \exp\left(-\frac{d(p, \mathcal{S}_T)^2}{2\sigma^2}\right) \right], \quad \sigma = 0.14. \quad (8)$$

The first factor penalizes missing target segments and the second penalizes extraneous generated segments. We sample 12 points per target segment and 6 points per generated segment.

Base prior. At $t = 0$, sequence length is uniform on $\{0, \dots, 12\}$ and strings are uniform conditional on length. This non-degenerate source ensures that the alignment baseline encounters insertion, deletion, and substitution; with an empty source its forward alignment contains only insertions. The base insertion, deletion, and substitution rates are 1.0, 0.4, and 0.8, respectively, and the terminal loss is $g = -r/0.025$. The exact target p_1^* is computed for this fixed source distribution using sparse matrix-exponential products rather than by applying a global terminal tilt to p_1^{base} .

MLPs. All neural methods use the same state representation and MLP trunk. A state is represented by a normalized 20×20 raster of its rendered curve together with its endpoint coordinates, heading $(\sin \phi, \cos \phi)$, and normalized sequence length; time is appended as an additional input. The network has two hidden layers of width 128 with SiLU activations and is optimized with Adam at learning rate 2×10^{-3} . DEAM uses $K = 8$ completion rollouts per state and minibatches of 32. The amortized variant uses $M = 8$ supervised neighbours by default, a delayed critic updated every 500 iterations, and holds the actor at the base model for the first 2000 iterations.

Metrics. We report terminal $D_{\text{KL}}(p_1^* \| p_1^u)$, with both marginals computed exactly on the enumerable state space, and the forward path KL to the discretized exact-adjoint reference u^{ref} . For two CTMCs sharing p_0 ,

$$D_{\text{KL}}(p^u \| \mathcal{P}^{\text{ref}}) = \int_0^1 \sum_x p_t^u(x) \sum_{y \neq x} \left[u_t(y | x) \log \frac{u_t(y | x)}{u_t^{\text{ref}}(y | x)} - u_t(y | x) + u_t^{\text{ref}}(y | x) \right] dt, \quad (9)$$

weighted by the model’s own occupancy p_t^u and evaluated on the same 20 piecewise-constant time bins as the learned controllers.

B.2 A2D2 conditional-path baselines

Both arms use A2D2’s trajectory-level endpoint weighting with the main distinction being the conditional path used for supervision.

Exact CTMC bridge. For an endpoint z , $p_{1|t}^{\text{base}}(z | x)$ is available exactly here by a sparse matrix exponential. The base CTMC conditioned on $X_1 = z$ has state distribution and rates

$$q_t^{\text{bridge}}(x | z) = \frac{p_t^{\text{base}}(x) p_{1|t}^{\text{base}}(z | x)}{p_1^{\text{base}}(z)}, \quad u_t^{\text{bridge}}(y | x; z) = u_t^{\text{base}}(y | x) \frac{p_{1|t}^{\text{base}}(z | y)}{p_{1|t}^{\text{base}}(z | x)}. \quad (10)$$

The endpoint weight targets $p_1^{\text{base}}(z)e^{-g(z)}$, cancelling the normalization of q_t^{bridge} , and $p_t^{\text{base}}(x)p_{1|t}^{\text{base}}(z | y)$ cancels in the resulting rate ratio, so the population fixed point is

$$u_t^{\text{base}}(y | x) \frac{\sum_z e^{-g(z)} p_{1|t}^{\text{base}}(z | y)}{\sum_z e^{-g(z)} p_{1|t}^{\text{base}}(z | x)} = u_t^*(y | x)$$

for any source p_0 . Trajectory-level credit is therefore consistent when paired with the base bridge. Computing that bridge is unavailable for a general pretrained Edit Flow, so this arm is an oracle diagnostic.

Alignment interpolant. A2D2-Align replaces the bridge by the conditional path used in Edit Flow training: given endpoints (x_0, z) we take a minimum-edit alignment, let each aligned column independently reach its target with probability $\kappa_t = t$, and assign rate $1/(1-t)$ to every edit that fixes a still-unfixed column and zero to all others. This transports the weighted endpoint marginals correctly, which is why the terminal KL improves, but the same computation leaves no dependence on u^{base} : the alignment path is an exogenous interpolant rather than a conditioning of p^{base} , so its fixed point is determined by the coupling alone and does not target (1).

B.3 Path-weight validation

The estimator in (4) rests on $\mathbb{E}_{p^u}[e^{W_{[t,1]}(X_{t:1})-g(X_1)} \mid X_t = x] = e^{-V_t(x)}$, and on the enumerable problem the right-hand side is available exactly from a sparse matrix exponential. We check it at the configuration of section 3 using the learned DEAM rate field from each of the three independently trained DEAM seeds on three probe states x : the empty string, a prefix of the target, and the target itself and evaluate at $t_0 \in \{0.1, 0.5, 0.9\}$ for each of the three seeds, with 4000 completions per cell (x, t_0) . Table 3 summarizes the outcome.

Table 3: Path-weight validation over 27 cells. *No exit* retains only the jump term in (3); *no jump* retains only the integrated exit-rate term.

Weight	median rel. dev.	median $ \log_{10} \text{ratio} $	max $ \log_{10} \text{ratio} $
Full $W_{[t,1]}$	3.1%	0.01	0.09
no exit	2885%	1.70	2.80
no jump	$1.2 \times 10^6\%$	4.07	10.02

The full weight is within 17.9% in all 27 cells while dropping either component instead produces errors of several orders of magnitude, confirming that both terms in (3) are necessary.

C Toy Experiment Results

C.1 Main Results

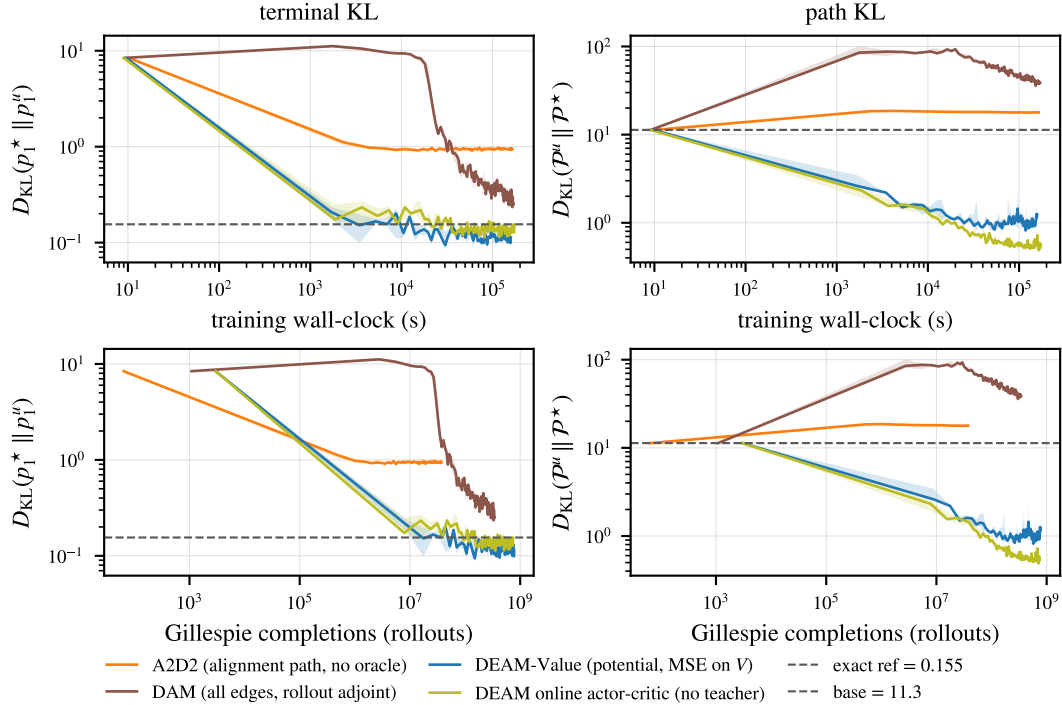


Figure 3: Terminal KL (left) and path KL (right) against training wall-clock (top) and rollout budget (bottom). Median over five seeds with an interquartile band. Per-rollout cost spans two orders of magnitude across methods, so the same 48 hours gives 4×10^7 Gillespie completions for A2D2 and 8×10^8 for DEAM. The ordering is unchanged on both axes.

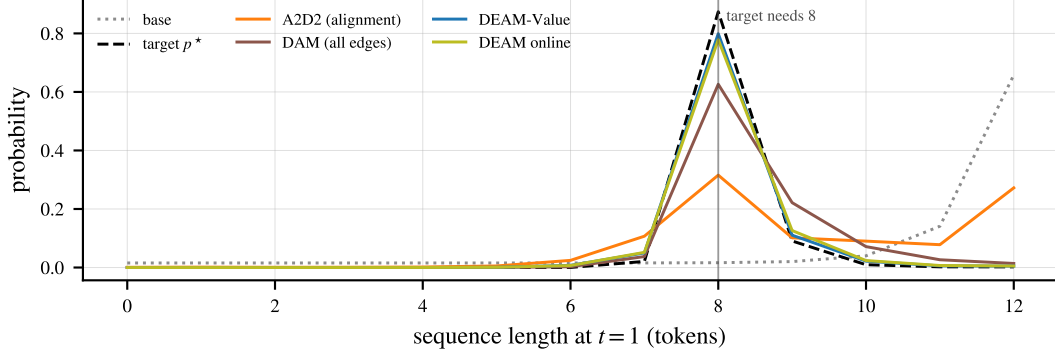


Figure 4: Distribution over sequence length at $t = 1$, averaged over seeds. The target zigzag needs 8 tokens. The base process is uniform over lengths, so every arm must steer length to match p^* .

C.2 Exit-rate Analysis

Path KL measures discrepancy from the optimal path law. To assess whether the learned dynamics perform too many or too few edits on average, we compare their expected total edit count with that of the optimum:

$$\bar{\Lambda}^u := \int_0^1 \sum_x p_t^u(x) \Lambda_t^u(x) dt, \quad (11)$$

Table 4: Mean exit rate at the matched 48-hour budget (5 seeds, mean $\pm$ sd), relative to the exact-adjoint reference.

Rate field	$\bar{\Lambda}^u$	$\bar{\Lambda}^u / \bar{\Lambda}^{u^*}$	path KL
Reference (u^*)	20.0	1.00	—
Base model	18.8	0.94	11.3
A2D2, alignment coupling	4.0 ± 0.1	0.20	17.8
DAM, all edges	61.1 ± 1.8	3.05	38.5
DEAM	19.7 ± 0.1	0.98	1.0
DEAM-online	19.9 ± 0.2	1.00	0.5

The optimum makes about as many edits as the base process (20.0 against 18.8). Both DEAM variants recover this to within 2% without being constrained to, because $s_\theta(x, y, t) = V_\theta(x, t) - V_\theta(y, t)$ ties every edge out of x to one scalar and is antisymmetric on reversible edits, so raising the outflow of x lowers the inflow by the same amount. Neither baseline has such a coupling: the free per-edge head runs $3.05\times$ too fast, and the alignment coupling $5\times$ too slow, since its interpolant makes only the minimal corrective edits while the base process wanders.

C.3 Supervised neighbours per state

Figure 5 sweeps the number of supervised neighbours M in (7) with the proposal ρ held fixed. At the matched 48-hour budget, terminal KL is 0.135 ± 0.028 , 0.128 ± 0.010 , 0.146 ± 0.010 , 0.133 ± 0.006 and 0.136 ± 0.009 for $M = 1, 2, 4, 8, 16$, and path KL is likewise flat (0.54 to 0.73). No setting is separated from the others by more than its own seed spread, so a single supervised neighbour per visited state already suffices at this edit degree.

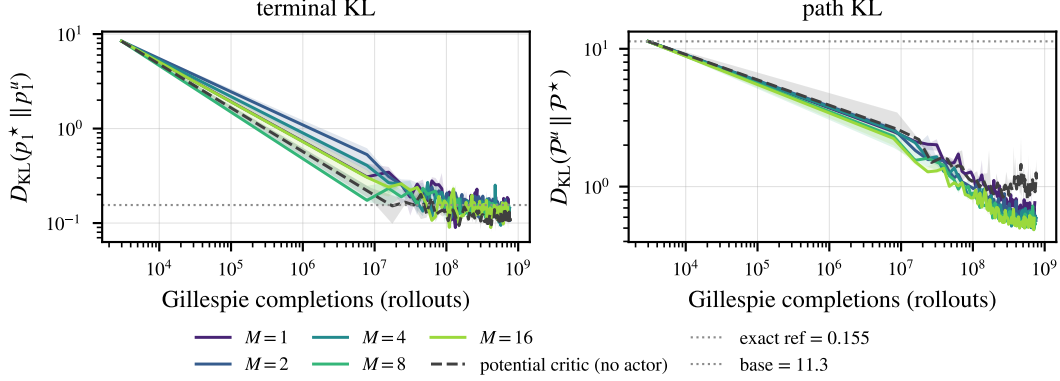


Figure 5: Terminal KL (left) and path KL (right) against rollout budget for $M \in \{1, 2, 4, 8, 16\}$ and the potential critic (dashed). Median over three seeds with an interquartile band; five seeds for $M = 8$ and the critic. Only one cost axis is shown because all arms are the same method at different settings.

C.4 Ablation: parametrization and target space

We separate two choices in DEAM: the parametrization of the correction and the space in which the noisy rollout target is matched. For a state x , let

$$\hat{h}_t(x) := \frac{1}{K} \sum_{k=1}^K \exp(W^{(k)} - g(Z^{(k)})), \quad \hat{V}_t(x) = -\log \hat{h}_t(x),$$

and for an edit $x \rightarrow y$ let $\hat{a}_t(x, y) := \frac{\hat{h}_t(y)}{\hat{h}_t(x)}$.

The free-rate parametrization predicts an independent log-rate correction $s(x, y, t)$ for each edit, whereas the potential parametrization predicts $\hat{V}_\theta(x, t)$ and uses $s(x, y, t) = \hat{V}_\theta(x, t) - \hat{V}_\theta(y, t)$.

For each parametrization we compare matching before and after the logarithm.

- The free-rate model either applies the generalized-KL rate loss to $\hat{a}$, as in DAM, or regresses s onto $\log \hat{a}$.
- The potential model either matches the unbiased estimator $\hat{h}$ through the corresponding generalized-KL loss, or regresses directly onto $\hat{V} = -\log \hat{h}$, which is the DEAM objective in (5).

All four arms use the same architecture, sampled states, and training budget.

Table 5: Parametrization and loss ablation at the configuration of Section 3 (5 seeds for the rate-gKL and DEAM rows, 3 elsewhere; mean $\pm$ sd). Structure is the reverse-edge ratio: 0 is exactly antisymmetric and 1 exactly symmetric.

Targets	Parametrization	Loss / target	$D_{\text{KL}}(p_i^* p_i^u)$	path KL	structure
rollout	free per-edge	rate gKL on $\hat{a}$	0.287 ± 0.029	38.5	0.938
rollout	free per-edge	MSE on $\log \hat{a}$	1.037 ± 0.039	15.0	0.989
rollout	scalar potential	gKL on $\hat{h}$	$3.783 \pm 0.135^\dagger$	7.9	—
rollout	scalar potential	MSE on $\hat{V}$ (DEAM)	0.114 ± 0.009	1.0	0.000
exact	free per-edge	rate gKL	0.160 ± 0.000	0.05	0.088
exact	free per-edge	log-space MSE	0.192 ± 0.001	0.10	0.084

No loss structure successfully adapts the free per-edge parametrization of DAM. Rate-space gKL reaches 0.287 and log-space regression 1.037, the latter despite consuming $4\times$ the rollouts. Imposing the scalar-potential structure lowers terminal KL to 0.114. Conversely, the potential alone is not sufficient: matching $\hat{h}$ directly diverges to a non-finite rate field in all three seeds. Both the potential

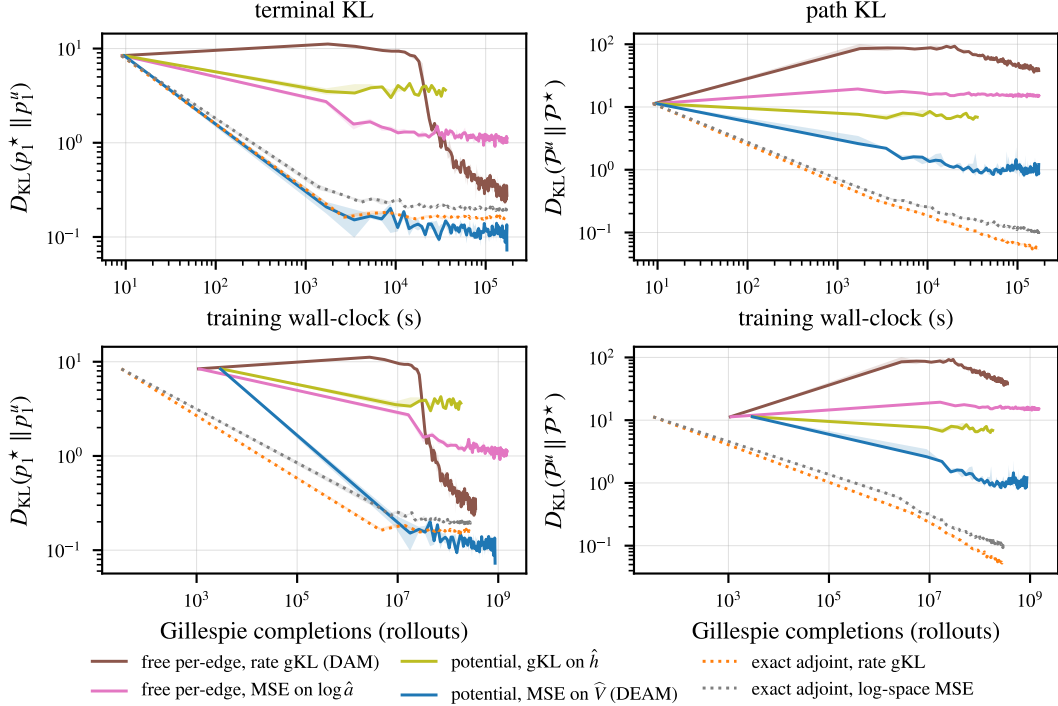


Figure 6: Terminal KL (left) and path KL (right) against training wall-clock (top) and rollout budget (bottom), for the four rollout-target ablations and the two exact-target controls (dotted). Median over seeds with an interquartile band; the gKL-on- $\hat{h}$ traces stop where each seed diverged.

structure and the log-space value regression are therefore required under finite-sample rollout targets. The free heads also fail to recover antisymmetry on reversible edits (structure 0.94 and 0.99, against 0 for a potential by construction), which is what allows them to inflate the total exit rate.

The exact-target rows are controls for the free per-edge head. Once rollout noise is removed, both losses approach the exact-adjoint reference and recover near antisymmetric corrections, showing that the failure above is driven by noisy targets rather than an inability of the rate head to represent the solution.

[†]All three seeds produced a non-finite rate field, after 2.8, 3.9 and 9.9 hours; the row reports the last five evaluations before divergence.

D Molecule experiment details

D.1 Setup

Base model. The pretrained any-length insertion MDM released with A2D2 [6]: 96M parameters over SAFE strings, vocabulary 1882, maximum length 256, sampled with 256 Euler steps. Mean generated length is ≈ 83 tokens, so a trajectory takes ≈ 166 jumps. Insertion places a mask rather than a token, so the marked edit degree is $Lq + L + 1 \approx 1.6 \times 10^5$ at that length, rising to 4.8×10^5 at the maximum length 256.

Architecture. Both actor and critic read the same frozen backbone. One forward pass on (x_t, t) yields per position features $h_0, \dots, h_L \in \mathbb{R}^{768}$: the L token positions followed by the terminal gap, which two independent heads then use. Let $c = (t, L/256, \text{mask fraction})$ be three scalar state features appended to every head input, and $\bar{h}$ the mean of h_j over those $L+1$ active positions.

- **Critic.** $V_\theta(x, t) = \psi(\bar{h}, c) + \sum_j \chi(h_j, c)$, with ψ a 3-layer MLP of width 256 on the pooled feature and χ a 2-layer MLP of width 128 applied to each h_j and summed. Mean-pooling alone divides a one-edit change by the number of active positions, attenuating it 7–14 $\times$ at typical lengths and suppressing exactly the differences (6) consumes; summing moves V_θ by the edited position’s own term at full magnitude.
- **Actor.** $s_\phi(x, y, t)$ emits every log-tilt in one pass. A token head maps the L token features to $\mathbb{R}^q$, giving $L \times q$ unmasking tilts; a gap head maps all $L+1$ features to $\mathbb{R}$, giving $L+1$ insertion tilts: insertion places a mask, so it carries no vocabulary factor. Both are 2-layer MLPs of width 128. Rates are nonzero only on masked positions and on gaps below the maximum length, so one pass covers the whole one-edit neighbourhood, of size $\approx Lq$.

Every head is LayerNorm $\rightarrow$ Linear $\rightarrow$ SiLU $\rightarrow \dots \rightarrow$ Linear with a zero-initialized output layer, so $V_\theta \equiv 0$ and $u^\phi = u^{\text{base}}$ at initialization: training starts exactly at the base model. Neither head backpropagates into the backbone; together they total $\approx 1\text{M}$ parameters against 96M frozen. Only the actor runs at sampling time: the critic supplies the actor’s targets during training and is then discarded.

V_α scales as $1/\alpha$ to leading order, so scaling the actor’s target by γ approximates the policy at temperature α/γ . This matters because a value head trained directly at small α is not estimable: its target contains $e^{r/\alpha}$, and at $\alpha = 0.0125$ a $K = 8$ log-mean-exp is dominated by the single largest completion. We estimate where the estimator is well conditioned and deploy where the reward is, using $\gamma = 12$. The approximation is first order and loosens as γ grows.

Reward. $r = \text{QED} + 1/(1 + \text{SA})$, the objective of Tang et al. [6]; QED and SA are the RDKit and Ertl and Schuffenhauer [2] implementations exposed by TDC. Invalid decodings score zero. We also report a QED-only ablation, $r = \text{QED}$.

Training. DEAM trains a value head and the amortized edit head of Section 2 ($\approx 1\text{M}$ parameters) in a single loop from random initialization; there is no pretrained critic. The actor is held at the base model for the first 600 updates while the value head warms up, which is the warm-up of Algorithm 1 rather than a separate stage. We use $\alpha = 0.1$, $K = 8$ completions, $M = 32$ supervised neighbours, batch 4, Adam at 3×10^{-4} for the actor and 3×10^{-5} for the value head, and a delayed critic refreshed every 20 updates.

Evaluation. 1000 samples $\times$ 3 seeds, decoded with the same SAFE-to-SMILES path as Tang et al. [6]. Every evaluation resamples the base model through our sampler under the same seed, so the reported difference is paired and not read across two sampling implementations.

Completions The value target $-\log \frac{1}{K} \sum_k e^{W_k + r_k/\alpha}$ requires completions on the base measure. Sampling them from the model and reweighting by W , as So et al. [5] do, is the lower-variance choice when the reward is a rare event: base rollouts then almost never reach the states the tilted distribution concerns, which is not the regime here. With $K = 8$ base completions the target’s estimator noise is 0.188 against a true target spread of 0.592, a reliability of 0.91.

At the temperature that maximizes reward the model-sampled weights instead degenerate and training runs diverge, with validity falling to 0.47–0.82, and this holds both for forward-simulated states and for states reciprocally projected through the base interpolant. We therefore take training states from model rollouts and completions from the base process, which removes the correction rather than approximating it.

D.2 Full results

Table 6 reports the metrics of Tang et al. [6]. Quality is the fraction of samples simultaneously valid, unique, drug-like (QED ≥ 0.6) and synthesizable (SA ≤ 4). It is gated on SA, which is why the QED-only ablation recovers A2D2’s QED yet reaches only two-thirds of its quality: optimizing QED alone drives SA past the gate, an effect Tang et al. [6] also report.

Table 6: Drug-like small molecule design, 1000 samples $\times$ 3 seeds. A2D2 rows from Tang et al. [6]. Base measured through our sampler.

Method	Validity	Uniqueness	Quality	QED ($\uparrow$)	SA ($\downarrow$)	Diversity
Pre-trained (any-length)	98.20	99.63	45.47	0.653	3.472	0.904
A2D2	94.53	93.70	71.30	0.762	2.870	0.843
DEAM (ours)	99.37	99.90	72.23	0.717	2.807	0.873
<i>DEAM, QED-only</i>	<i>99.40</i>	<i>99.97</i>	<i>47.33</i>	<i>0.755</i>	<i>3.924</i>	<i>0.891</i>